FROM MARBLE TO LIME: ARCHAEOMETRIC AND EXPERIMENTAL INVESTIGATION OF HISTORICAL LIME PRODUCTION IN RORÀ, WALDENSIAN VALLEYS, NW ITALY

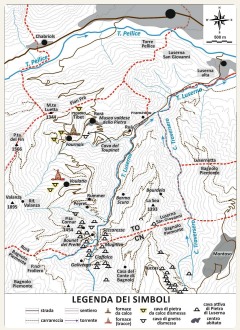
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1 CONTEXT AND RESEARCH QUESTION

A local stone for a long lime-burning tradition

- Lime production at Rorà is documented since 1251.
- Small carbonate lenses occur within the Dora-Maira metamorphic sequence.
- The inhabitants were known as brusapère, “stone burners”.



Rorà district and historical lime-kiln sites. Modified from Vola (2025).

2 GEOLOGICAL SETTING

Carbonate lenses

- Typically 5–10 m thick and laterally discontinuous.
- Interlayered with garnet micaschists, fine-grained gneisses and Pietra di Luserna orthogneiss.
- Dolomitic marbles pass locally into impurity-rich calcschists.

Age interpretation

Probably Mesozoic cover tectonically interleaved within the Paleozoic basement. This remains a multi-criterion interpretation, not a direct age determination.



Fournais: historical kiln and nearby carbonate outcrops.

3 MATERIALS AND INTEGRATED APPROACH

13 carbonate-rock samples

5 sampling sites

From rock to lime performance

1. **FIELD** — Geological survey and historical evidence
2. **ROCK** — PLM · CLM · XRPD-QPA · WD-XRF · ICP-MS
3. **SIGNATURE** — C–O isotopes · EPR · SEM-EMPA
4. **FIRING** — Calcination Tests at 1050–1150 °C
5. **PERFORMANCE** — Available Lime Index (ALI); mechanical degradation (MD); drop test (DT); and overburning test (OBT) at 1300 °C

4 MINERALOGICAL AND PETROGRAPHIC RESULTS

A heterogeneous dolomitic system

DOLOMITIC MARBLES

dominant lithology

INSOLUBLE RESIDUE

mean 3.6 wt%, max 8.5 wt%

CALCSCHISTS

locally up to 66 wt% non-carbonates

Texture

Predominantly fine-grained and compositionally variable. The Rorà materials mainly lack the relic high-pressure pyroxenes found in some Paleozoic Dora-Maira marbles.

5 GEOCHEMICAL AND ISOTOPIC CONSTRAINTS

Main marble cluster

0 to +1.5‰ $\delta^{13}C$ (VPDB)

−6 to −8.5‰ $\delta^{18}O$ (VPDB)

Outliers record localized re-equilibration

Calcschists reach $\delta^{13}C \approx -3.4$‰. RIO-1 has $\delta^{18}O = -13.5$‰, consistent with stronger fluid–rock interaction or higher-temperature re-equilibration.

6 ARCHAEOMETRIC EVIDENCE

Residues preserve the firing history

- Periclase records dolomite decomposition.
- Brucite and hydromagnesite record hydration and natural recarbonation.
- Larnite may form at standard calcination temperatures in impure rocks where free silica is available. Alite and vitrified slags point to locally higher temperatures.

7 TECHNOLOGICAL BEHAVIOUR

Most samples are unsuitable for modern vertical kilns

12.8% MD, TIB-1

57–79% MD, 5 of 6 samples

Mechanical degradation (MD)

Five samples produced excessive fines, threatening bed permeability and gas flow.

Sticking tendency at 1300 °C

MOR-1, TIB-1, VOL-1 and VOL-2 showed low values (5–10 g). Impure FOR-4 and FOR-5 underwent partial melting and reached 100% sticking.

Key point: fragmentation and sticking measure different failure modes and must be assessed jointly.

8 WHY DID HISTORICAL PRODUCTION WORK?

Traditional intermittent kilns could accommodate raw-material variability

- **Manual selection** — Workers selected the most suitable carbonate levels.
- **Small batches** — Operators adjusted firing duration and intensity.
- **Wood firing** — Lower and less uniform thermal loads reduced industrial-style constraints.
- **Product sorting** — Post-firing selection compensated for variable lime quality.

9 CONCLUSIONS

RORÀ MARBLES DEFINE A DISTINCT LOCAL CARBONATE RESOURCE

- Fine-grained dolomitic marbles and calcschists form a strongly heterogeneous system.
- Field relations, petrography and isotopes support a probable Mesozoic-cover affinity.
- Impurities and microstructure control fragmentation, sticking and lime reactivity.
- The same variability that limits modern continuous kilns remained manageable in small historical furnaces.
- Residues and slags provide direct material evidence of past lime burning.

Historical success reflects skilled adaptation to a difficult raw material.

INTEGRATED INTERPRETATION

GEOLOGY	RAW MATERIAL	FIRING RESPONSE	HISTORICAL CHOICE
Meta-carbonate lenses interlayered with para and orthogneisses of the Dora-Maira Paleozoic basement (Cottian Alps, NW Italy)	Mainly dolomitic marbles, calcschists, and sporadic calcitic marbles	Reactivity may be high, but mechanical degradation and sticking tendency vary independently	Flexible intermittent kilns made selective local use possible

THE ARCHAEOLOGICAL RECORD AND THE EXPERIMENTAL TESTS TELL THE SAME STORY

Rorà lime production succeeded through geological knowledge, selective quarrying and adaptable firing practice. The resource was locally valuable, but it does not meet the uniformity demanded by modern continuous kilns.

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Selected references: Vola et al. (2022), J. Archaeol. Sci. Rep. 45, 103562 · Geological and analytical data: Vola et al., in preparation.